Worldline instantons in an “inflation” metric: the $\beta = 0$ starting point

Independent reproduction of Sec. III A–B of P. Semrén and G. Torgrimsson, “Worldline instantons for nonperturbative particle production by space and time dependent gravitational fields”, arXiv:2508.01901.

The pair-production probability is P ~ exp(–$\mathcal{A}$). Its exponent follows from a saddle point of the worldline path integral: a geodesic in complex proper time u, called a worldline instanton. The complex middle part of the instanton describes the formation (tunnelling) region; its two ends become the real trajectories of the produced particle and antiparticle.

Metric (Eq. 21): $ds^2 = dt^2 - e^{2A(t,x)}\, dx^2$, with $A = \alpha \tanh(\omega t)\, \mathrm{sech}^2(kx)$ (Eq. 24).
Here $\beta = k/\omega = 0$, so the metric depends on t only. In units $\omega = 1$: $A(t) = \alpha \tanh t$.

$\alpha = 1$ and p = 0.57 (close to the saddle-point momentum p_s) reproduce Fig. 2 of the paper.
The complex u-plane is covered by $-2 < \mathrm{Re}\, u < 2$ and $-4 < \mathrm{Im}\, u < 4$.
Exact rational inputs are used so that NDSolve can work at the requested precision.

The instanton starts at the branch point t_B (Eq. 27). For $\beta = 0$, Eq. (22) integrates once to (Eq. 25)
$\dot{x} = p\, e^{-2A(t)}$, $\dot{t} = \sqrt{1 + p^2\, e^{-2A(t)}}$.
The middle of the instanton is the turning point $\dot{t} = 0$, i.e. $e^{2A(t_B)} = -p^2$, which gives
$t_B = \mathrm{arctanh}[(i\pi + \ln p^2)/(2\alpha)]$.

```
ClearAll["Global`*"];

tInterp = Import[saveDir <> "tInterp.wdx"];

alpha = SetPrecision[1, 50];
p = SetPrecision[57 / 100, 50];

Rmin = -4;
Rmax = 4;

urMin = -2;
urMax = 2;

xdot[R_?NumericQ, ur_?NumericQ] := p * Exp[-2 * alpha * Tanh[tInterp[R, ur]]];
```

Geodesic equations (Eq. 22), with dots denoting d/du:
$\ddot{t} = (1 - \dot{t}^2)\, \partial_t A$, $\ddot{x} = -2\, \dot{t}\, \dot{x}\, \partial_t A - \dot{x}^2\, \partial_x A$.
The first form follows from $\Gamma^t_{xx} = \tfrac{1}{2}\, \partial_t e^{2A}$ and the on-shell condition $\dot{t}^2 - e^{2A}\, \dot{x}^2 = 1$ (Eq. 8).
For $\beta = 0$, $\partial_x A = 0$.

Initial conditions at u = 0: t = t_B, $\dot{t} = 0$, x = 0, $\dot{x}$ = p e^{–2A(t_B)}.

The solution is first obtained along the imaginary axis u = iR (the “backbone”).
Since d/du = –i d/dR, with T(R) = t(iR) and X(R) = x(iR):
T’’ = –(1 + T’^{2}) ∂_t A(T), X’’ = –2 T’ X’ ∂_t A(T), X’(0) = i $\dot{x}$(0).

solving eqn 22

In[•]:=
```
A[t_] := alpha * Tanh[t];
At[t_] := alpha * Sech[t]^2;

tB = ArcTanh[(I * Pi + Log[p^2]) / (2 * alpha)];


Rstep = 0.02;
urStep = 0.01;

RGrid = Range[Rmin, Rmax, Rstep];
urGrid = Range[urMin, urMax, urStep];
```

In[•]:=
```
solBackbone = NDSolve[
   {T''[R] == -(1 + T'[R]^2) At[T[R]], X''[R] == -2 T'[R] × X'[R] × At[T[R]], T[0] == tB,
    T'[0] == 0, X[0] == 0, X'[0] == I * p * Exp[-2 * A[tB]]}, {T, X}, {R, Rmin, Rmax},
   WorkingPrecision → 50, AccuracyGoal → 30, PrecisionGoal → 30, MaxSteps → Infinity];

Tfun[R_?NumericQ] := Evaluate[T[R] /. solBackbone[[1]]];

Xfun[R_?NumericQ] := Evaluate[X[R] /. solBackbone[[1]]];

dTfun[R_?NumericQ] := Evaluate[T'[R] /. solBackbone[[1]]];

dXfun[R_?NumericQ] := Evaluate[X'[R] /. solBackbone[[1]]];
```

NDSolve: The precision of the differential equation
({{T′′[R] == 1.000000000000000000000000000000000000000000000000 Sech[T[R]]^2 (–1 – ≪1≫$^{(≪1≫)}$[≪1≫]^2), X′′[R] == –2`.
.000000000000000000000000000000000000000000000000 Sech[T[R]]^2 T′[R] X′[R], T[0] == –0.1531963232248`.
1153428569187014609280529313239442 85 + 1.043551136431598870876835144216887958609273346129 1 *i*
, T′[0] == 0, X[0] == 0, X′[0] == 0. × 10^{-49} – 1.754385964912280701754385964912280701754385964912 *i*}, {}, {}, {},
{}}) is less than WorkingPrecision (50.`).

In[•]:=
```
horizontalData = Table[Module[{t0, dt0, x0, dx0, sol}, t0 = Tfun[R0];
    dt0 = -I * dTfun[R0];
    x0 = Xfun[R0];
    dx0 = -I * dXfun[R0];
    sol = NDSolve[
      {t''[ur] == (1 - t'[ur]^2) At[t[ur]], x''[ur] == -2 t'[ur] × x'[ur] × At[t[ur]],
       t[0] == t0, t'[0] == dt0, x[0] == x0, x'[0] == dx0}, {t, x}, {ur, urMin, urMax},
      WorkingPrecision → 40, AccuracyGoal → 20, PrecisionGoal → 20, MaxSteps → Infinity];
    Table[{ur, R0, Evaluate[t[ur] /. sol⟦1⟧], Evaluate[x[ur] /. sol⟦1⟧], Evaluate[
       t'[ur] /. sol⟦1⟧], Evaluate[x'[ur] /. sol⟦1⟧]}, {ur, urGrid}]], {R0, RGrid}];

flatData = Flatten[horizontalData, 1];

tGrid = Partition[flatData⟦All, 3⟧, Length[urGrid]];
tdotGrid = Partition[flatData⟦All, 5⟧, Length[urGrid]]
xGrid2 = Partition[flatData⟦All, 4⟧, Length[urGrid]];
xdotGrid = Partition[flatData⟦All, 6⟧, Length[urGrid]]
tInterp2 =
  ListInterpolation[tGrid, {{Rmin, Rmax}, {urMin, urMax}}, InterpolationOrder → 1];
tdotInterp2 =
  ListInterpolation[tdotGrid, {{Rmin, Rmax}, {urMin, urMax}}, InterpolationOrder → 1];
xInterp2 =
  ListInterpolation[xGrid2, {{Rmin, Rmax}, {urMin, urMax}}, InterpolationOrder → 1];
xdotInterp2 =
  ListInterpolation[xdotGrid, {{Rmin, Rmax}, {urMin, urMax}}, InterpolationOrder → 1];
```

Out[•]=
```
{{-1.02144 + 0.00294933 i, -1.02143 + 0.00300994 i, -1.02142 + 0.0030718 i, -1.02141 + 0.00313492 i,
  -1.0214 + 0.00319933 i, -1.02139 + 0.00326505 i, -1.02138 + 0.00333213 i, -1.02137 + 0.00340057 i,
  -1.02136 + 0.00347041 i, -1.02135 + 0.00354168 i, -1.02134 + 0.00361441 i, -1.02133 + 0.00368863 i, ··· 378 ···,
  -1.84542 + 0.00024601 i, -1.84538 + 0.00023707 i, -1.84533 + 0.000228455 i, -1.84528 + 0.000220155 i,
  -1.84524 + 0.000212157 i, -1.8452 + 0.000204451 i, -1.84516 + 0.000197025 i, -1.84512 + 0.00018987 i,
  -1.84509 + 0.000182976 i, -1.84505 + 0.000176332 i, -1.84501 + 0.000169931 i}, ··· 399 ···, {··· 1 ···}}
```

Full expression not available (original memory size: 10.3 MB)

Out[•]=
```
{{0.0760198 - 0.0105704 i, 0.0759883 - 0.0107876 i, 0.0759558 - 0.0110091 i, 0.0759223 - 0.0112352 i,
  0.0758877 - 0.011466 i, 0.075852 - 0.0117014 i, 0.0758151 - 0.0119417 i, 0.0757771 - 0.0121869 i,
  0.0757378 - 0.012437 i, 0.0756973 - 0.0126923 i, 0.0756554 - 0.0129528 i, 0.0756122 - 0.0132186 i,
  ··· 378 ···, 4.22033 - 0.00159296 i, 4.22002 - 0.00153503 i, 4.21972 - 0.00147921 i, 4.21943 - 0.00142543 i,
  4.21915 - 0.00137362 i, 4.21889 - 0.0013237 i, 4.21863 - 0.00127559 i, 4.21838 - 0.00122924 i,
  4.21814 - 0.00118458 i, 4.21791 - 0.00114155 i, 4.21768 - 0.00110009 i}, ··· 399 ···, {··· 1 ···}}
```

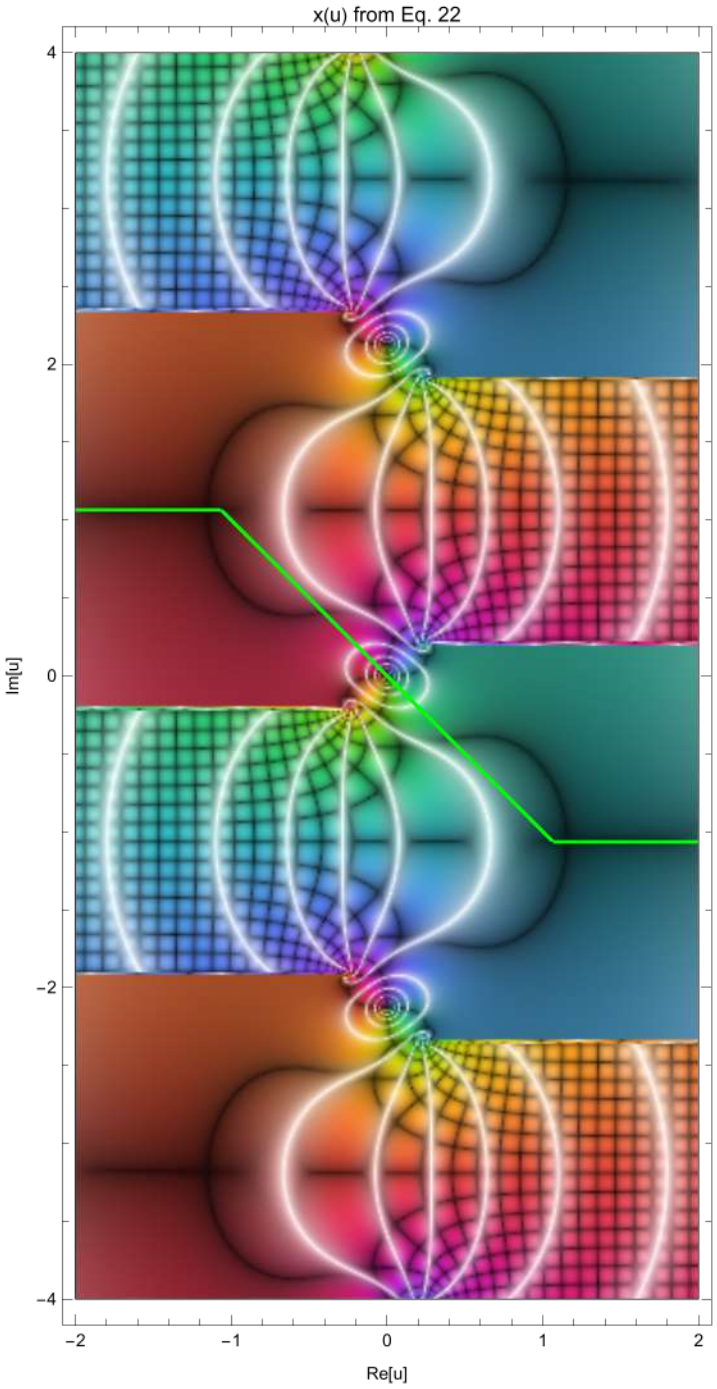
Full expression not available (original memory size: 10.3 MB)

In[•]:=
```
Rc = 1.06595;

greenContour = Graphics[{Directive[Green, Thick], (*upper horizontal*)
     Line[{{-2, Rc}, {-Rc, Rc}}], (*diagonal*)Line[{{-Rc, Rc}, {Rc, -Rc}}],
     (*lower horizontal*)Line[{{Rc, -Rc}, {2, -Rc}}]}];

x = Show[ComplexPlot[xInterp2[Im[u], Re[u]],
    {u, urMin + I * Rmin, urMax + I * Rmax}, ColorFunction → "CyclicReImLogAbs",
    PlotPoints → 120, MaxRecursion → 3, ImageSize → Large,
    FrameLabel → {"Re[u]", "Im[u]"}, PlotLabel → "x(u) from Eq. 22"], greenContour]
```

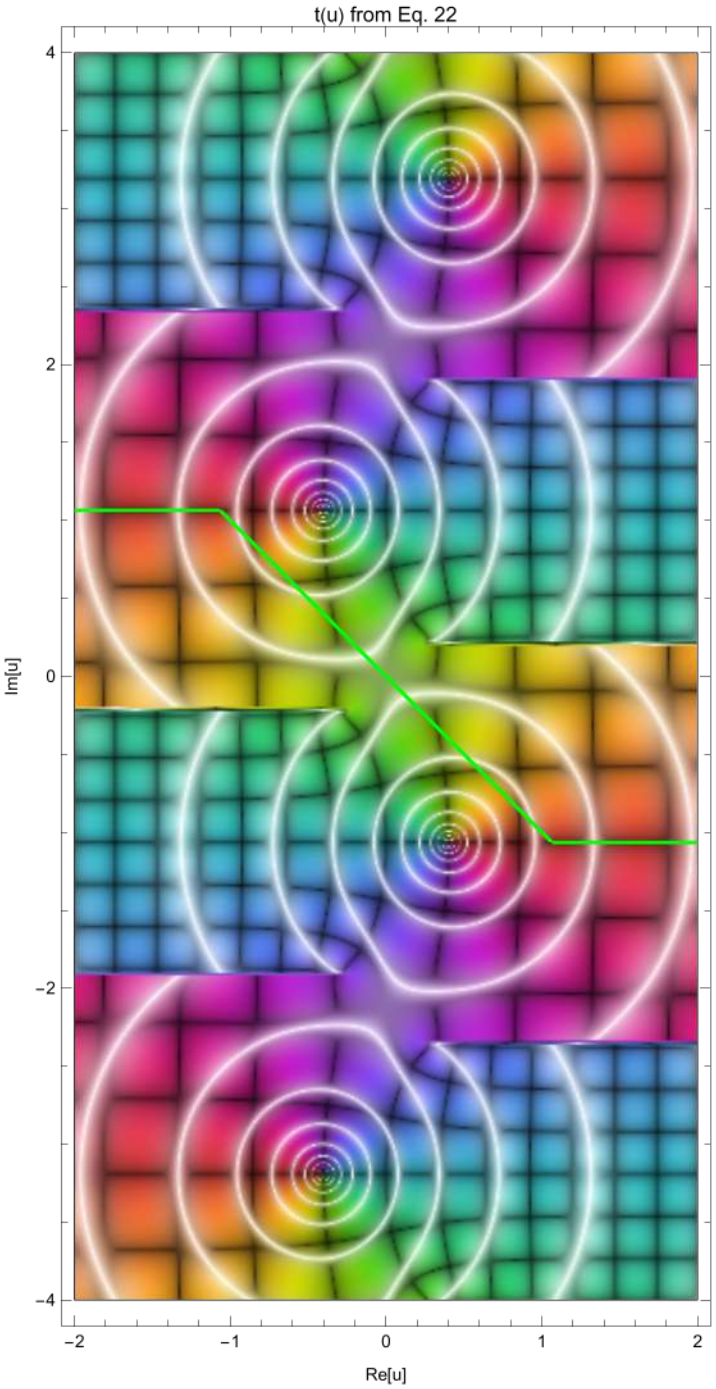
Out[ ]=



In[•]:=

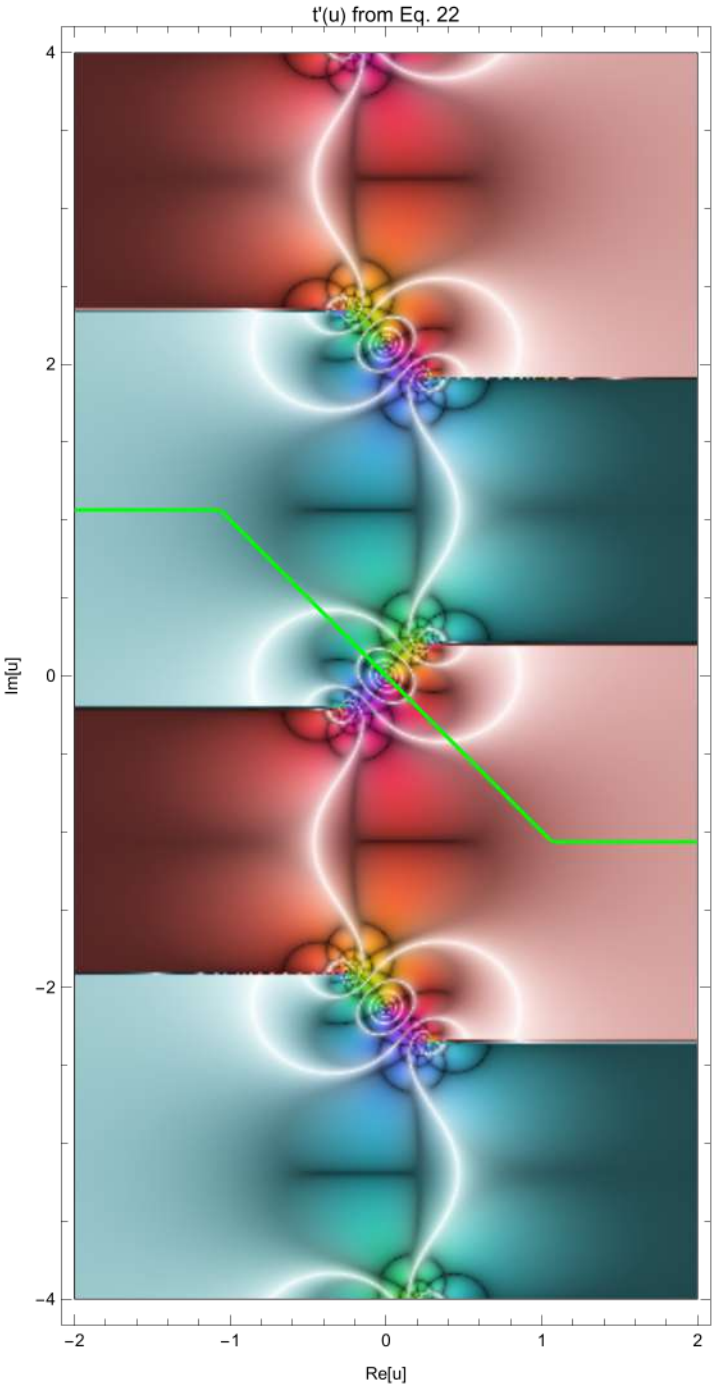
```
t = Show[ComplexPlot[tInterp2[Im[u], Re[u]],
    {u, urMin + I * Rmin, urMax + I * Rmax}, ColorFunction → "CyclicReImLogAbs",
    PlotPoints → 120, MaxRecursion → 3, ImageSize → Large,
    FrameLabel → {"Re[u]", "Im[u]"}, PlotLabel → "t(u) from Eq. 22"], greenContour]
```

Out[•]=



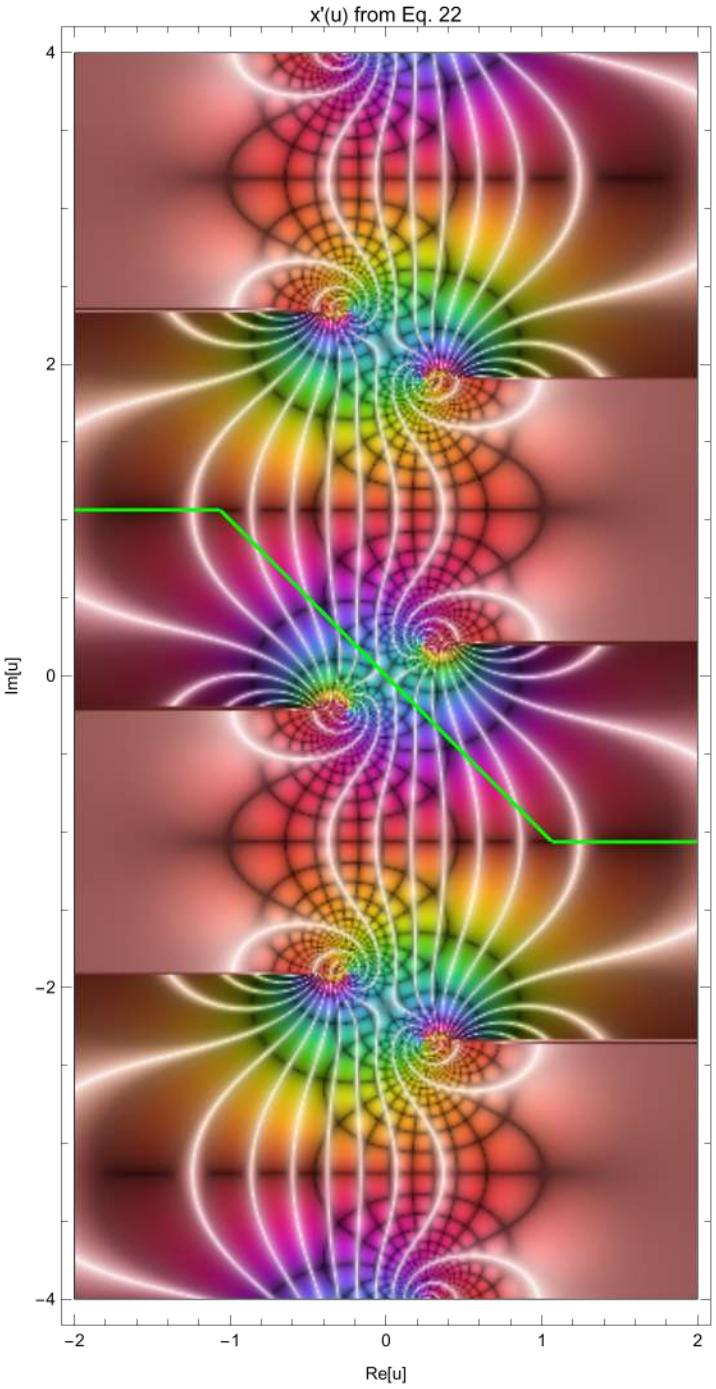
In[•]:=
```
tprime = Show[ComplexPlot[tdotInterp2[Im[u], Re[u]],
    {u, urMin + I * Rmin, urMax + I * Rmax}, ColorFunction → "CyclicReImLogAbs",
    PlotPoints → 120, MaxRecursion → 3, ImageSize → Large,
    FrameLabel → {"Re[u]", "Im[u]"}, PlotLabel → "t'(u) from Eq. 22"], greenContour]
```

Out[•]=



```
In[•]:= xprime = Show[ComplexPlot[xdotInterp2[Im[u], Re[u]],
    {u, urMin + I * Rmin, urMax + I * Rmax}, ColorFunction → "CyclicReImLogAbs",
    PlotPoints → 120, MaxRecursion → 3, ImageSize → Large,
    FrameLabel → {"Re[u]", "Im[u]"}, PlotLabel → "x'(u) from Eq. 22"], greenContour]
```

Out[•]=



The instanton must become real at both ends, so that it describes real asymptotic particles. At the saddle-point momenta this requires (Eq. 20)

Im t = Im x = Im $\dot{t}$ = Im $\dot{x}$ = 0.

On the imaginary axis, t(u) becomes real at u = iRc. Since Eq. (22) has real coefficients, t, x, $\dot{t}$, $\dot{x}$ then stay real along the whole horizontal line Im u = Rc, including the corner u = –Rc + iRc.

Contour (green line): horizontal for Re u < –Rc, diagonal from –Rc + iRc to Rc – iRc, then horizontal again. Beyond the corners the solution stays real, so the integration can stop there.

Checks:
(i) integrating straight along the diagonal to the corner gives vanishing imaginary parts (Eq. 20);
(ii) the diagonal route and the grid route give the same values at the corner, so no singularity lies between them and the contour is unambiguous.
Small residuals (~10^{-4}) come from p = 0.57 being only approximately p_s, and from the linear interpolation.

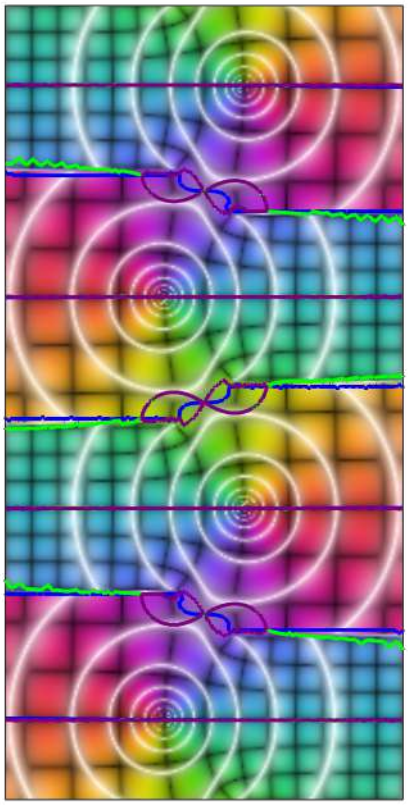
green contour selection

In[•]:=
```
cp = ComplexPlot[tInterp2[Im[u], Re[u]],
    {u, urMin + I * Rmin, urMax + I * Rmax}, ColorFunction → "CyclicReImLogAbs",
    PlotPoints → 120, MaxRecursion → 3, Frame → False];

contours = Show[ContourPlot[Im[tInterp2[R, ur]], {ur, urMin, urMax}, {R, Rmin, Rmax},
     Contours → {0}, ContourShading → None, ContourStyle → {Red, Thick}, Frame → False],
    ContourPlot[Im[xInterp2[R, ur]], {ur, urMin, urMax}, {R, Rmin, Rmax},
     Contours → {0}, ContourShading → None, ContourStyle → {Blue, Thick}, Frame → False],
    ContourPlot[Im[tdotInterp2[R, ur]], {ur, urMin, urMax}, {R, Rmin, Rmax},
     Contours → {0}, ContourShading → None, ContourStyle → {Green, Thick}, Frame → False],
    ContourPlot[Im[xdotInterp2[R, ur]], {ur, urMin, urMax}, {R, Rmin, Rmax}, Contours →
      {0}, ContourShading → None, ContourStyle → {Purple, Thick}, Frame → False]];

Show[cp, contours, PlotRange → All]
```

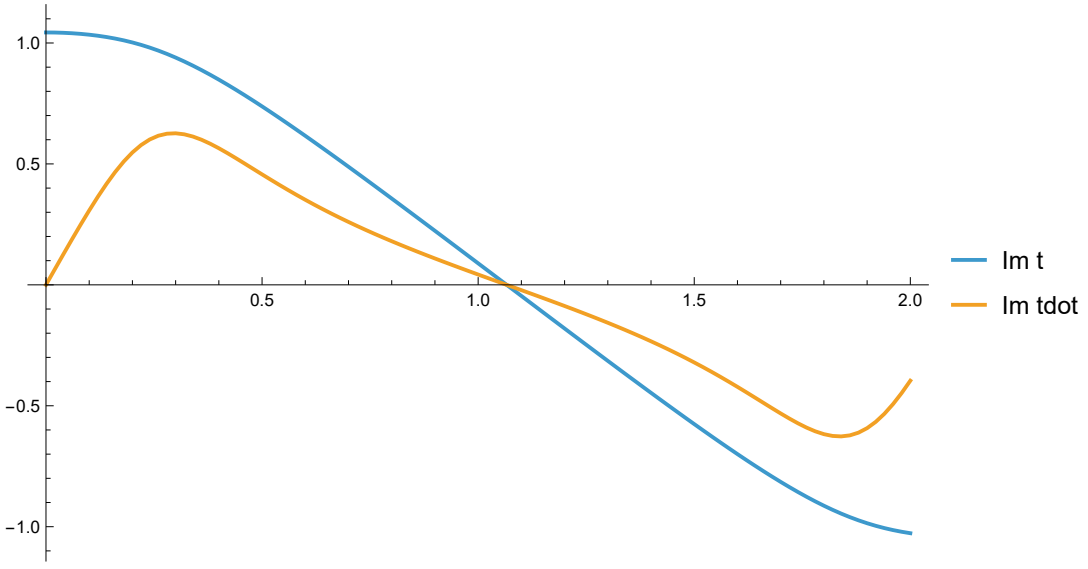
Out[•]=



In[•]:=
```
upperData = Table[{R, Im[tInterp2[R, 0]], Im[tdotInterp2[R, 0]]}, {R, 0, 2, 0.01}];
Select[upperData, Abs[#[[2]]] < 10^-2 && Abs[#[[3]]] < 10^-2 &][[All, 1]]
```

Out[•]=
```
{1.06, 1.07}
```

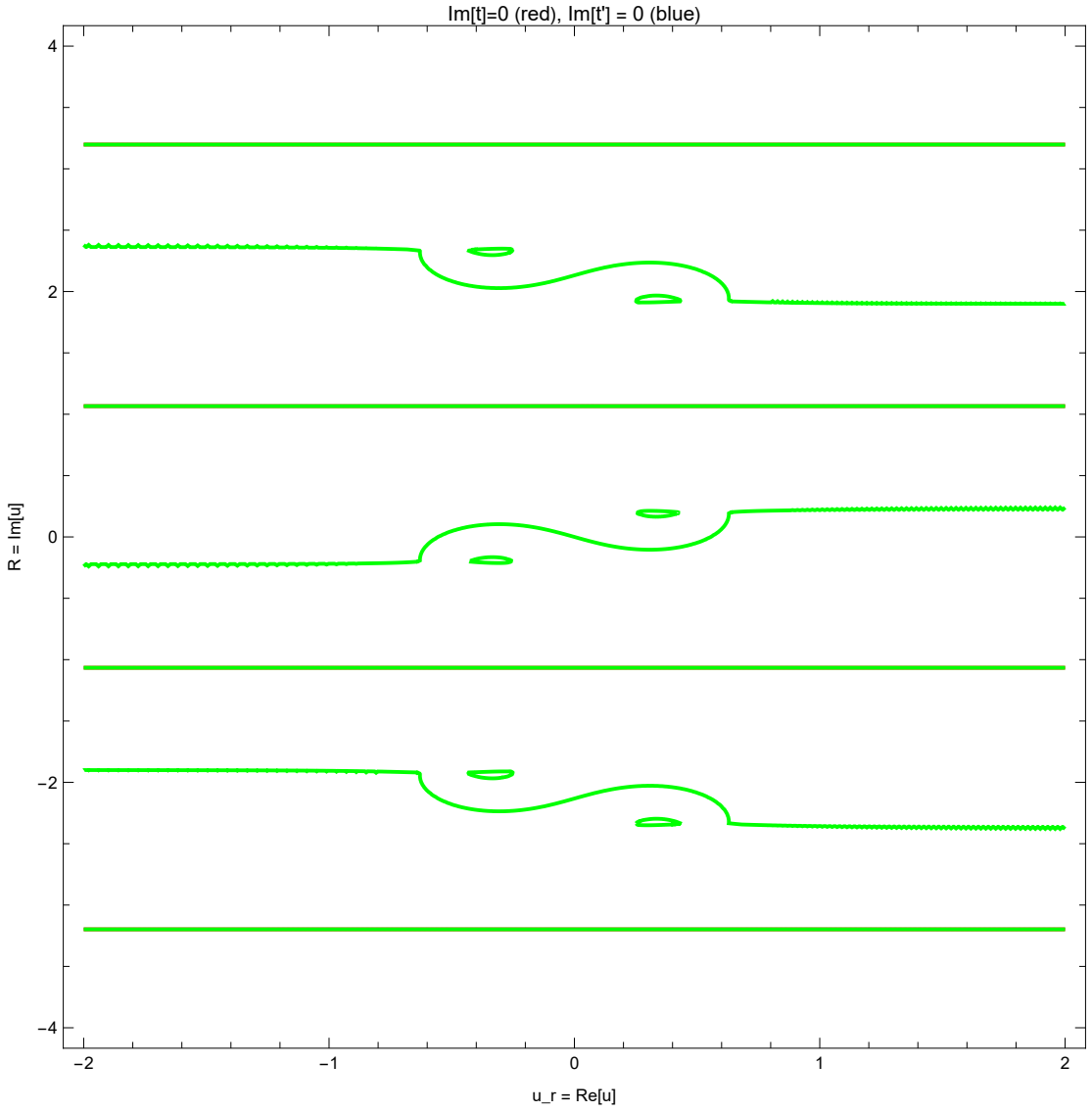
Out[•]=



```
In[•]:= upperBranch =
          Table[{ur, Im[tInterp2[1.07, ur]], Im[tdotInterp2[1.07, ur]]}, {ur, -2, 0, 0.001}];
        zeros = Select[upperBranch, Abs[#[[2]]] < 10^-2 && Abs[#[[3]]] < 10^-2 &];
        zeros[[All, 1]]

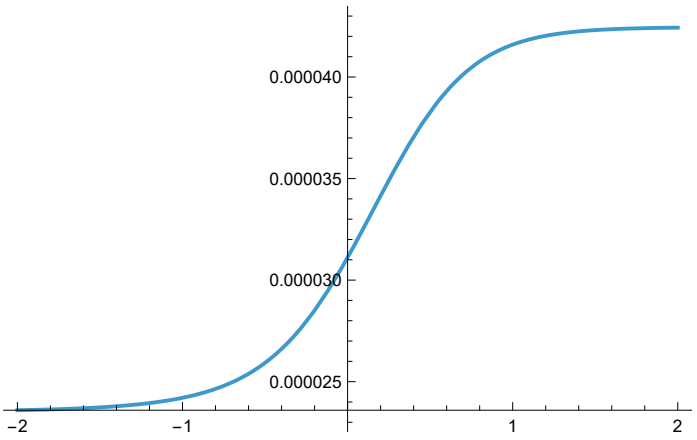
In[•]:= Show[ContourPlot[Im[tInterp2[R, ur]], {ur, urMin, urMax}, {R, Rmin, Rmax},
          Contours → {0}, ContourShading → None, ContourStyle → {Red, Thick},
          PlotPoints → 100, MaxRecursion → 2], ContourPlot[Im[tdotInterp2[R, ur]],
          {ur, urMin, urMax}, {R, Rmin, Rmax}, Contours → {0}, ContourShading → None,
          ContourStyle → {Green, Thick}, PlotPoints → 100, MaxRecursion → 2],
         Frame → True, FrameLabel → {"u_r = Re[u]", "R = Im[u]"},
         PlotLabel → "Im[t]=0 (red), Im[t'] = 0 (blue)", ImageSize → Large]
```

Out[•]=



In[•]:=
```
Plot[Im[tInterp2[1.06595, ur]], {ur, -2, 2}]
```

Out[•]=



t(u), x(u), ṫ(u), ẋ(u) in the complex proper-time plane, plotted with ColorFunction → “CyclicReImLogAbs” (compare Fig. 2 of the paper). Red means real and positive, blue real and negative; black lines are lines of constant real or imaginary part.

The Im = 0 curves of t, x, ṫ, ẋ are overlaid in the last plot: the contour turns horizontal where they intersect.

In[•]:= **GraphicsRow[{t, x, tprime, xprime}, ImageSize → 1300]**

Out[•]=

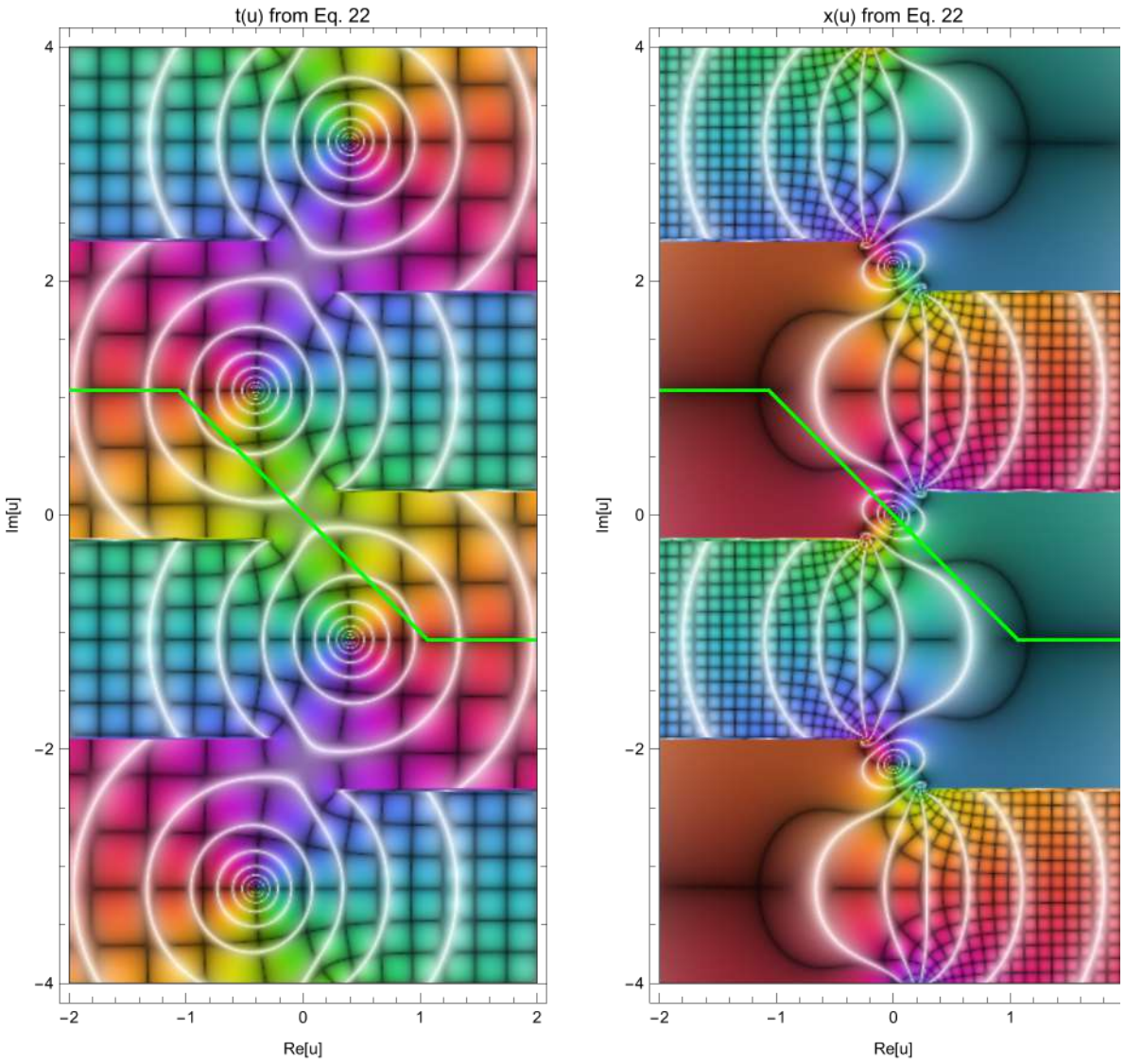


figure 3

In[•]:=
```
Rc = 1.06595;
uContour[r_] :=
  Piecewise[{{r + I Rc, -6 ≤ r ≤ -Rc}, {r - I r, -Rc ≤ r ≤ Rc}, {r - I Rc, Rc ≤ r ≤ 6}}];

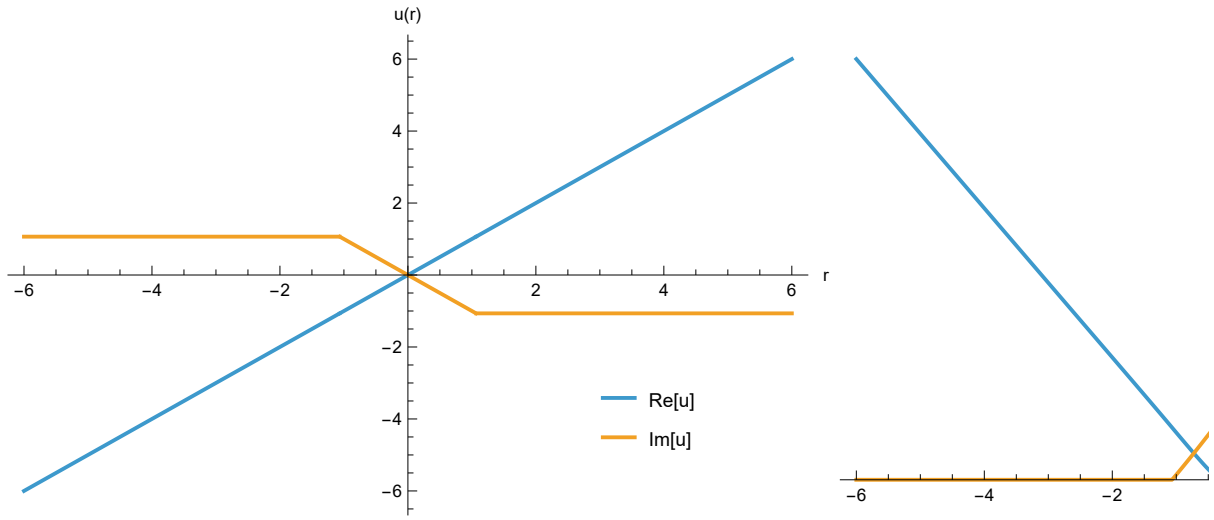
rVals = Range[-6, 6, 0.01];

fig3Data = Table[{r, Re[tInterp2[Im[uContour[r]], Re[uContour[r]]]],
    Im[tInterp2[Im[uContour[r]], Re[uContour[r]]]],
    Re[xInterp2[Im[uContour[r]], Re[uContour[r]]]],
    Im[xInterp2[Im[uContour[r]], Re[uContour[r]]]],
    Re[tdotInterp2[Im[uContour[r]], Re[uContour[r]]]],
    Im[tdotInterp2[Im[uContour[r]], Re[uContour[r]]]],
    Re[xdotInterp2[Im[uContour[r]], Re[uContour[r]]]],
    Im[xdotInterp2[Im[uContour[r]], Re[uContour[r]]]]}, {r, rVals}];
```

In[•]:=
```
uPlot = Plot[{Re[uContour[r]], Im[uContour[r]]},
   {r, -6, 6}, PlotLegends → Placed[{"Re[u]", "Im[u]"}, {0.8, 0.2}],
   AxesLabel → {"r", "u(r)"}, AspectRatio → 0.6, ImageSize → 500];
tPlot = ListLinePlot[{fig3Data[[All, {1, 2}]], fig3Data[[All, {1, 3}]]},
   PlotLegends → Placed[{"Re[t]", "Im[t]"}, {0.8, 0.2}],
   AxesLabel → {"r", "t(r)"}, AspectRatio → 0.6, ImageSize → 500];
xPlot = ListLinePlot[{fig3Data[[All, {1, 4}]], fig3Data[[All, {1, 5}]]},
   PlotLegends → Placed[{"Re[x]", "Im[x]"}, {0.8, 0.8}],
   AxesLabel → {"r", "x(r)"}, AspectRatio → 0.6, ImageSize → 500];

GraphicsGrid[{{uPlot, tPlot, xPlot}}, ImageSize → 1300, Spacings → {0.5, 0.5}]
tdotPlot = ListLinePlot[
   {fig3Data[[All, {1, 6}]], fig3Data[[All, {1, 7}]]}, PlotLegends → {"Re[t']", "Im[t']"},
   AxesLabel → {"r", "t'(r)"}, AspectRatio → 0.6, ImageSize → 500];
xdotPlot = ListLinePlot[
   {fig3Data[[All, {1, 8}]], fig3Data[[All, {1, 9}]]}, PlotLegends → {"Re[x']", "Im[x']"},
   AxesLabel → {"r", "x'(r)"}, AspectRatio → 0.6, ImageSize → 500];
GraphicsGrid[{{tdotPlot, xdotPlot}}, ImageSize → 1300, Spacings → {0.5, 0.5}]
```

Out[•]=



In[•]:=
```
0.57 * Exp[-2]
```
Out[•]=
```
0.0771411
```

In[•]:=
```
Re[xdotInterp2[-Rc, 2]]
```
Out[•]=
```
0.0881837
```

In[•]:=
```
Re[p Exp[-2 A[tInterp2[-Rc, 2]]]]
```
Out[•]=
```
0.0881859
```